6 The curve C has polar equation $r = \sin 3\theta$, for $0 \leqslant \theta \leqslant \frac{1}{3}\pi$.

(a) Sketch C and state the equation of the line of symmetry. [3]

.....

(b) Find the exact value of the area of the region enclosed by C. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

In parts **(c)** and **(d)** you may use the identity $\sin 3\theta \equiv 3\sin\theta - 4\sin^3\theta$.

(c) Find the maximum distance of a point on C from the initial line. [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(d) Find a Cartesian equation for C. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....